IMPLEMENTAREA CONCURENTEI IN LIMBAJE DE PROGRAMARE

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INTRODUCERE IN ERLANG

http://www.erlang.org/

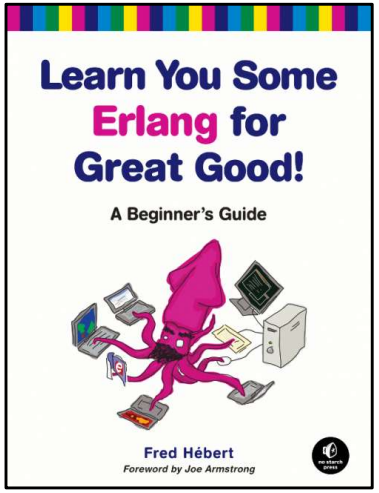
ACTOR MODEL

"Erlang's actor model can be imagined as a world where everyone is sitting alone in their own room and can perform a few distinct tasks. Everyone communicates strictly by writing letters and that's it. While it sounds like a boring life (and a new age for the postal service), it means you can ask many people to perform very specific tasks for you, and none of them will ever do something wrong or make mistakes which will have repercussions on the work of others; they may not even know the existence of people other than you (and that's great).

To escape this analogy, Erlang forces you to write actors (processes) that will share no information with other bits of code unless they pass messages to each other. Every communication is explicit, traceable and safe."

Fred Hébert, Learn You Some Erlang For Great Good

http://learnyousomeerlang.com/introduction#what-is-erlang



Varianta online

➢Bibliografie

Joe Armstrong, Robert Virding, Mike Williams, Concurrent Programming in Erlang, 1993

Joe Armstrong, Programming Erlang, Second Edition 2013

Fred Hébert, Learn You Some Erlang For Great Good, 2013

https://www.erlang.org/doc/

CONCURRENCY IN ERLANG

lightweight processes with asynchronous message passing

Procesele in Erlang:

- pot fi create si distruse rapid
- comunica prin mesaje, iar comunicarea este rapida
- sunt complet independente din punctul de vedere al memoriei

- Concurenta in Erlang este implementata folosind urmatoarele primitive:

```
Pid = spawn (fun)

Pid = spawn (module, fct, args)

Pid ! Message

receive … end
```

https://www.erlang.org/doc/man/erlang.html#spawn-4

"Messages between Erlang processes are simply valid Erlang terms. That is, they can be lists, tuples, integers, atoms, pids, and so on.

Each process has its own input queue for messages it receives. New messages received are put at the end of the queue. When a process executes a receive, the first message in the queue is matched against the first pattern in the receive. If this matches, the message is removed from the queue and the actions corresponding to the pattern are executed.

However, if the first pattern does not match, the second pattern is tested. If this matches, the message is removed from the queue and the actions corresponding to the second pattern are executed. If the second pattern does not match, the third is tried and so on until there are no more patterns to test. If there are no more patterns to test, the first message is kept in the queue and the second message is tried instead. If this matches any pattern, the appropriate actions are executed and the second message is removed from the queue (keeping the first message and any other messages in the queue). If the second message does not match, the third message is tried, and so on, until the end of the queue is reached. If the end of the queue is reached, the process blocks (stops execution) and waits until a new message is received and this procedure is repeated."

http://erlang.org/doc/getting_started/conc_prog.html

➢Modelul Actori

- Introdus de Carl Hewitt in 1973
- Actorii sunt o notiune abstracta (corespunzatoare proceselor)
- Actorii au memorie proprie, NU au memorie partajata
- Actorii comunica prin mesaje
- Un actor este capabil sa:
    - trimita mesaje actorilor pe care ii cunoaste
    - creeze noi actori
    - raspunda mesajelor pe care le primeste
- Mesajele contin un destinatar si un continut
- Trimiterea mesajelor este asincrona

➢ Cilent-Server (Exemplu simplu: doubling service)

```
3> c(myserv).
{ok,myserv}
4> Ser=spawn(myserv, server_loop, []).
<0.44.0>
5> Ser ! {self(),{double,5}}.
{<0.32.0>,{double,5}}
6> flush().
Shell got {<0.44.0>,10}
ok
7> Ser ! {self(),{double,7}}.
{<0.32.0>,{double,7}}
8> flush().
Shell got {<0.44.0>,14}
ok
9> Ser ! {self(),111}.
{<0.32.0>,111}
10> flush().
Shell got {<0.44.0>,error}
ok
```

- Procesul **Ser** este serverul si executa functia **server_loop**
- Serverul primeste mesaje de la procese client si executa o actiune (dubleaza valoarea primita)
- In acest exemplu singurul client este shell-ul
- Mesajele primite de shell, adica raspunsurile trimise de server, sunt vizualizate cu **flush()**

➢ Cilent-Server (Exemplu simplu: doubling service)

```
-module(myserv).
-export([server_loop/0]).

server_loop() ->
  receive
      {From, {double, Number}} -> From ! {self(),Number*2},
                                  server_loop() ;

       {From,_} -> From ! {self(),error},
                   server_loop()
  end.
```

```
3> c(myserv).
{ok,myserv}
4> Ser=spawn(myserv, server_loop, []).
<0.44.0>
5> Ser ! {self(),{double,5}}.
{<0.32.0>,{double,5}}
6> flush().
Shell got {<0.44.0>,10}
ok
7> Ser ! {self(),{double,7}}.
{<0.32.0>,{double,7}}
8> flush().
Shell got {<0.44.0>,14}
ok
9> Ser ! {self(),lll}.
{<0.32.0>,lll}
10> flush().
Shell got {<0.44.0>,error}
ok
```

➢ Cilent-Server : functie pentru pornirea server-ului

```
-module(myserv).
-export([server_loop/0]).

server_loop() ->
  receive
      {From, {double, Number}} -> From ! {self(), Number*2},
                                  server_loop() ;

       {From,_} -> From ! {self(),error},
                   server_loop()
  end.
```

```
3> c(myserv).
{ok,myserv}
4> Ser=spawn(myserv, server_loop, []).
<0.44.0>
5> Ser ! {self(),{double,5}}.
{<0.32.0>,{double,5}}
6> flush().
Shell got {<0.44.0>,10}
ok
7> Ser ! {self(),{double,7}}.
{<0.32.0>,{double,7}}
8> flush().
Shell got {<0.44.0>,14}
ok
9> Ser ! {self(),111}.
{<0.32.0>,111}
10> flush().
Shell got {<0.44.0>,error}
ok
```

```
-export([start_server/0, server_loop/0]).
start_server() ->  spawn(myserv, server_loop, []).
```

```
16> Ser=myserv:start_server().
<0.66.0>
17> Ser ! {self(), {double,45}}.
{<0.59.0>,{double,45}}
18> flush().
Shell got {<0.66.0>,90}
ok
```

➢ Cilent-Server: functia client

```
-module(myserv).
-export([start_server/0, server_loop/0,client/2]).


start_server() ->  spawn(myserv, server_loop, []).


server_loop() ->
  receive
      {From, {double, Number}} -> From ! {self(),(Number*2)},
                                  server_loop() ;

      {From,_} -> From ! {self(),error},
                  server_loop()
  end.
```

```
client(Pid, Request) ->
      Pid ! {self(), Request},
      receive
            {Pid, Response} -> Response
       end.
```

functia **client intoarce raspunsul primit de la server**

```
3> c(myserv).
{ok,myserv}
4> Server = myserv:start_server().
<0.43.0>
5> myserv:client(Server,{double,15675}).
31350
6> myserv:client(Server,nothing).
error
7> myserv:client(Server, {double, 887}).
1774
```

apelarea functiei client

- Client-Server

  client_loop creaza mai multe procese client si intoarce lista rezultatelor

```
client_loop(Pid,0,L)  -> Pid! {self(),"Good Bye"},
                         L;

client_loop(Pid, X, L)  ->  R= client(Pid,{double,X}),
                            client_loop(Pid, X-1, L++[R]).
```

```
31> c(myserv2).
{ok,myserv2}
32> Ser = myserv2:start_server().
<0.113.0>
33> myserv2:client_loop(Ser,10,[]).
[20,18,16,14,12,10,8,6,4,2]
34> flush().
Shell got {<0.113.0>,"Good Bye"}
ok
```

- Client-Server

  client_loop creaza mai multi clienti si intoarce lista rezultatelor

```
client_loop(Pid,0,L)  -> Pid! {self(),"Good Bye"},
                         L;

client_loop(Pid, X, L)  ->  R= client(Pid,{double,X}),
                            io:fwrite("prel ~w!~n", [X]),
                            client_loop(Pid, X-1, L++[R]).
```

Functiile client sunt executate **secvential!**

➢ Client-Server

client_loop creaza mai multi clienti si intoarce lista rezultatelor

```
client_loop(Pid,0,L)  -> Pid! {self(),"Good Bye"},
                         L;

client_loop(Pid, X, L)  ->  R= client(Pid,{double,X}),
                            io:fwrite("prel ~w!~n", [X]),

                            client_loop(Pid, X-1, L++[R]).
```

```
28> Ser6=myserv2:start_server().
<0.139.0>
29> [myserv2:client_loop(Ser6, 3, []),myserv2:client_loop(Ser6, 3, [])].
prel 3!
prel 2!
prel 1!
prel 3!
prel 2!
prel 1!
```

Functiile client sunt executate **secvential!**

➢ Client-Server

client_loop creaza mai multi clienti si intoarce lista rezultatelor

```
client_loop(Pid,0,L)  -> Pid! {self(),"Good Bye"},
                         L;

client_loop(Pid, X, L)  ->  R= client(Pid,{double,X}),
                            io:fwrite("prel ~w!~n", [X]),

                            client_loop(Pid, X-1, L++[R]).
```

```
24> Ser5=myserv2:start_server().
<0.126.0>
25> [spawn(myserv2, client_loop, [Ser5, 3, []]),spawn(myserv2,client_loop,[Ser5, 3, []])].
prel 3!
prel 3!
[<0.128.0>,<0.129.0>]
prel 2!
prel 2!
prel 1!
prel 1!
```

Functiile client sunt executate **in paralel!**

➢Client-Server

procesele client se executa in **paralel** si se intoarce lista rezultatelor

```
worker(Parent, Pid, Number) ->   spawn( fun() ->
                                        Result = client (Pid,{double,Number}),
                                        Parent ! {self(),Result}
                                        end ).
calls (Pid,N) ->  Parent = self(),
                  Pids = [worker(Parent,Pid, X)  || X <- lists:seq(1,N)],
                  [ wait_one(P)  ||  P <- Pids ].

wait_one (Pid) ->
                 receive
                    {Pid,Response} -> Response
                 end.
```

% **Pid** este id-ul procesului server
% **Parent** este id-ul procesului care creaza clientii
% **worker** creaza un proces client si intoarce id-ul acestuia

- Client-Server

```
start_server() ->  spawn(myserv, server_loop, []).
start_seq_clients(Pid, N) -> client_loop(Pid,N,[]).
start_par_clients(Pid, N) -> calls(Pid,N).
```

```
62> c(myserv2).
{ok,myserv2}
63> Server = myserv2:start_server().
<0.15705.27>
64> myserv2:start_par_clients(Server, 100000).
[2,4,6,8,10,12,14,16,18,20,22,24,26,28,30,32,34,36,38,40,42,
 44,46,48,50,52,54,56,58|...]
65> myserv2:start_seq_clients(Server, 100000).
```

➢Client-Server

unui proces i se poate asocia un nume (atom) folosind register

myserv3.erl

```
start_server() ->  register(serv, spawn(fun() ->server_loop() end)).

server_loop() ->
  receive
    {From, {double, Number}} -> From ! {serv,(Number*2)},
                                server_loop() ;
      {From, "Good Bye"} -> From! {serv,"Good Bye"},
                              server_loop();

    {From,_} -> From ! {serv,error},
                server_loop()
  end.
```

procesul server are numele **serv**

- Registered process

register (Name, Pid) asociaza numele **Name** procesului cu id-ul **Pid**
Name este atom si este "eliberat" cand procesul se termina

whereis(Name) intoarce id-ul procesului inregistrat cu **Name**
(sau **undefined** daca nu exista)

Processes — Erlang System Documentation v27.2

```
start_server() ->  register(serv, spawn(fun() ->server_loop() end)).
```

➢ Client-Server

```
start_par_clients(N) -> calls(N).
worker(Parent,  Number) ->
   spawn( fun() ->
                 Result = client ({double,Number}),
                  Parent ! {self(),Result}
                end ).

calls (N) ->
      Parent = self(),
      Pids = [worker(Parent,X)|| X <- lists:seq(1,N)],
      [waitone(P)|| P <- Pids].



waitone (Pid) ->
                receive
                    {Pid,Response} -> Response
                 end.
```

```
client(Request) ->
        serv ! {self(), Request},
        receive
            {serv, Response} -> Response
        end.
```

```
1> cd ("D:/DIR/ER/myer").
D:/DIR/ER/myer
ok
2> c(myserv3).
{ok,myserv3}
3> myserv3:start_server().
true
4> myserv3:start_par_clients(50).
[2,4,6,8,10,12,14,16,18,20,22,24,26,28,30,32,34,36,38,40,42,
 44,46,48,50,52,54,56,58|...]
5>
```

Distributed Erlang: programele ruleaza in noduri diferite

```
PS C:\Users\igleu> cd C:\Users\igleu\Documents\DIR\ICLP\ICLP2024\c11-12-2024-Erlang\ER\myer
PS C:\Users\igleu\Documents\DIR\ICLP\ICLP2024\c11-12-2024-Erlang\ER\myer> erl -sname ioserver
Eshell V11.0  (abort with ^G)
(ioserver@LAPTOP-KCKGLLT6)1> myserv3:start_server().
true
```

>erl –sname ioserver

Command Prompt - erl -snan

```
Microsoft Windows [Version 10.0.22631.3593]
(c) Microsoft Corporation. All rights reserved.

C:\Users\igleu>erl -sname ioclient
Eshell V11.0  (abort with ^G)
(ioclient@LAPTOP-KCKGLLT6)1> {serv,'ioserver@LAPTOP-KCKGLLT6'}!{self(),{double,5}}.
{<0.84.0>,{double,5}}
(ioclient@LAPTOP-KCKGLLT6)2> flush().
Shell got {serv,10}
ok
(ioclient@LAPTOP-KCKGLLT6)3>
```

>erl –sname ioclient

Crearea unui nod in Erlang

>erl –sname …

Numele e atom!

Distributed Erlang: programele ruleaza in noduri diferite

erl -sname server

```
D:\DIR\ER\myer>erl -sname server
Eshell V7.3  (abort with ^G)
(server@Ioana-PC)1> myserv3:start_server().
true
(server@Ioana-PC)2>
```

Crearea unui nod in Erlang

>erl –sname …

erl -sname client

```
D:\DIR\ER\myer>erl -sname client
Eshell V7.3  (abort with ^G)
(client@Ioana-PC)1> rpc:call('server@Ioana-PC',myserv3, start_par_clients, [10]).
[2,4,6,8,10,12,14,16,18,20]
(client@Ioana-PC)2>
```

rpc:call
remote procedure call
http://erlang.org/doc/man/rpc.html#call-4

Distributed Erlang: programele ruleaza in noduri diferite

```
C:\Users\igleu>cd C:\Users\igleu\Documents\DIR\ICLP\ICLP2024\c11-12-2024-Erlang\ER\myer

C:\Users\igleu\Documents\DIR\ICLP\ICLP2024\c11-12-2024-Erlang\ER\myer>erl -sname ioserv
Eshell V11.0  (abort with ^G)
(ioserv@LAPTOP-KCKGLLT6)1>
```

Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvem

PS C:\Users\igleu> erl -sname new
Eshell V11.0  (abort with ^G)
(new@LAPTOP-KCKGLLT6)1>  spawn('ioserv@LAPTOP-KCKGLLT6', myserv3, start_server, []).
<8381.89.0>
(new@LAPTOP-KCKGLLT6)2>
```

cu spawn/4 serverul este pornit din alt nod
https://www.erlang.org/doc/apps/erts/erlang#spawn/4

Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\igleu> erl -sname ioclient
Eshell V11.0  (abort with ^G)
(ioclient@LAPTOP-KCKGLLT6)1> {serv,'ioserv@LAPTOP-KCKGLLT6'}!{self(),{double,5}}.
{<0.84.0>,{double,5}}
(ioclient@LAPTOP-KCKGLLT6)2> flush().
Shell got {serv,10}
```

➢ ?MODULE

macro (definit de sistem) care intoarce numele modulului curent

```
start() -> spawn(?MODULE, myrec, []).

myrec() ->
  receive
   {do_A, X} ->  prelA(X);
   {do_B, X} ->  prelB(X);
              _ -> io:format("Nothing to do ~n")
  end.
```

```
3> Pid = myconc1:start().
<0.112.0>
4> Pid ! {do_A,2}.
A
{do_A,2}
A
End A
```

➢ Cilent-Server (simple) template

```
-module(servtemplate).
-compile(export_all).

start_server() ->  register(serv,spawn(?MODULE, server_loop, [])).
```

```
client(Request) ->
        serv ! {self(), Request},
        receive
                {serv, Response} -> Response
         end.
```

```
server_loop() ->
   receive
        .....
        {From, Request} -> From ! {serv, Response} ,
                                server_loop()

        end.
```

tail recursion

- Concurenta in Erlang este implementata folosind urmatoarele primitive:

```
Pid = spawn (fun)

Pid = spawn (module, fct, args)

Pid ! Message

receive … end
receive … after … end
```

➢ receive … after … end

```
receive
Pattern1 when Guard1 -> Expr1;
Pattern2  when Guard2 -> Expr2;
Pattern3 -> Expr3
…
after T ->
      ExpressionT
end
```

- La intrarea in **receive**, daca exista un **after**, se porneste un timer.
- Mesajele din coada sunt investigate in ordinea sosirii; daca un mesaj se potriveste cu un pattern atunci expresia corespunzatoare este prelucrata.
- Mesajele care nu se potrivesc cu nici un pattern sunt puse intr-o coada separate (*save queue*).
- Daca nu mai sunt mesaje in coada procesul se suspenda si asteapta venirea unui nou mesaj; la venirea acestuia, numai el este prelucrat, nu si mesajele din *save queue.*
- Cand un mesaj se potriveste cu un pattern, mesajele din *save queue* sunt puse la loc in coada si timerul se sterge.
- Daca timpul T s-a scurs fara ca un mesaj sa se potriveasca unui pattern, atunci ExpressionT se executa, iar mesajele din *save queue* sunt puse inapoi in coada.

➢receive … after … end

```
sleep(T) ->
      receive
         after T ->
               ok
      end.
```

- nu exista sabloane, deci niciun mesaj nu va fi acceptat;
- procesul va fi blocat T milisecunde

```
flush() ->
  receive
              _      -> flush()
    after 0      ->
          ok
end.
```

- orice mesaj se potriveste cu patternul, deci apelul recursiv va goli coada de mesaje, dupa care procesul va continua

- instructiunea `after 0 -> …` verifica coada de mesaje si apoi continua; daca aceasta clauza lipseste, procesul se va bloca cand coada de mesaje se goleste

- flush()

```
flush() ->
  receive
        _        -> flush()
   after 0       ->
        ok
end.
```

```
5> self()! hi, self() ! low.
low
6> flush().
Shell got hi
Shell got low
ok
```

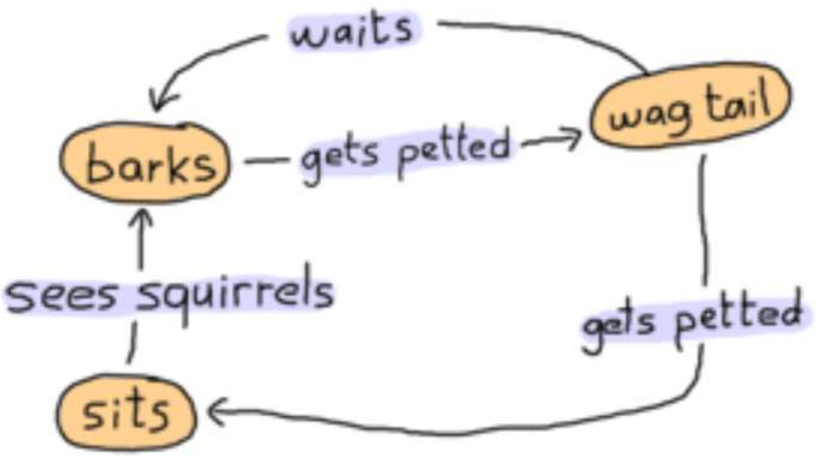
- receive cu prioritate

```
important() ->
    receive
      {Priority, X} when Priority > 10 ->   [X|important()]
     after 0 ->
            normal()
    end.

normal() ->
     receive
         {_,X} ->    [X|normal()]
      after 0 -> []
end.
```

Varianta a functiei flush() care ordoneaza mesajele dupa prioritati

```
2> c(sel).
{ok,sel}
3> self()! {5, low1}, self() ! {9,low2}, self() ! {15, high1}, self()!{11,high2}
.
{11,high2}
4> sel:important().
[high1,high2,low1,low2]
```

- Implementarea unui automat finit (Finite-State Machine)

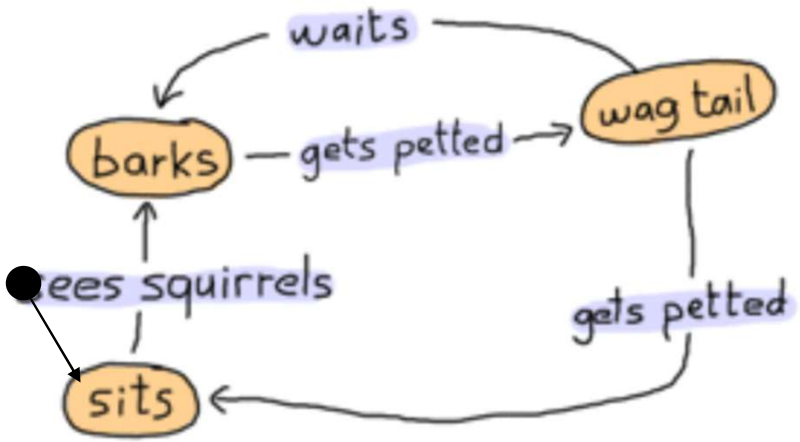


Starile ={barks, sits, wag_tail}
Actiunile ={gets_petted, see_squirrels, waits}

dog as a state-machine
http://learnyousomeerlang.com/finite-state-machines#what-are-they

➢ Implementarea unui automat finit

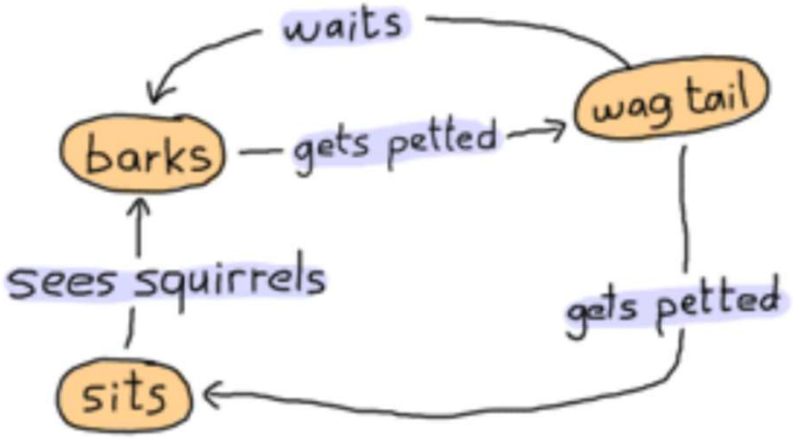
```
-module(dog_fsm).
-export([start/0, squirrel/1, pet/1]).

start() ->
    spawn(fun() -> bark() end).    % starea initiala

%actiunea  see_squirrels
squirrel(Pid) -> Pid ! squirrel.

%actiunea  gets_petted
pet(Pid) -> Pid ! pet.
```



actiunile sunt implementate prin mesaje si sunt vizibile in exterior

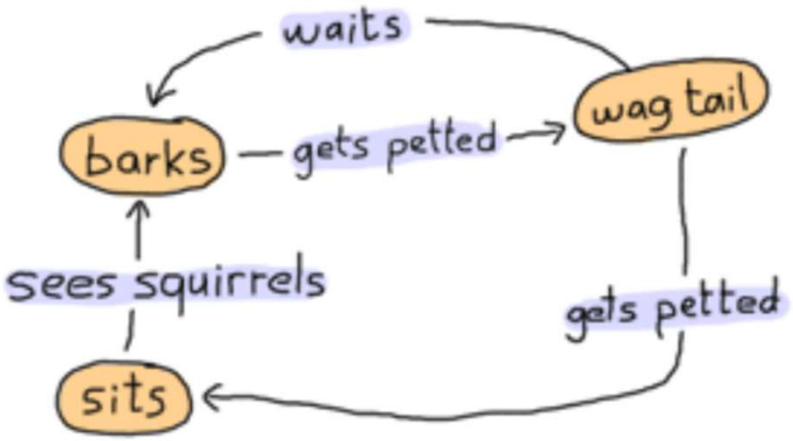
➢Finite-StateMachine: implementarea starilor

```
bark() ->
    io:format("Dog says: BARK! BARK!~n"),
   receive
     pet ->  wag_tail();

      _ ->  io:format("Dog is confused~n"),
            bark()

  after 2000 ->  bark()
end.
```



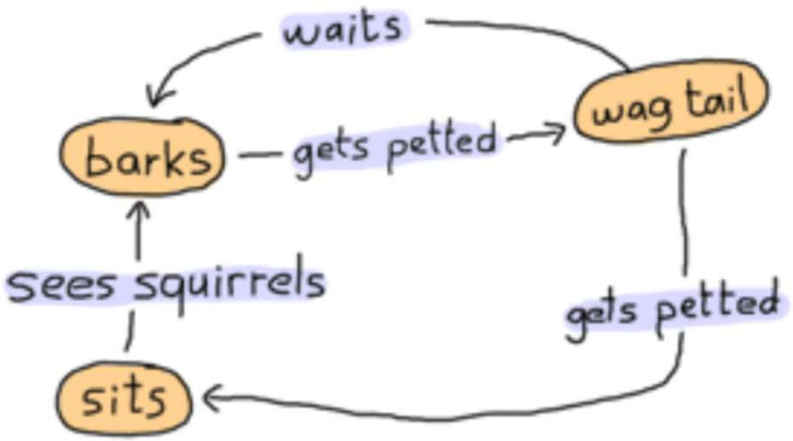
➢ Automat finit: definirea starilor

```
wag_tail() ->
        io:format("Dog wags its tail~n"),
        receive
            pet ->  sit();

            _ ->  io:format("Dog is confused~n"),
                  wag_tail()

       after 30000 ->
                    bark()      % actiunea waits

       end.
```



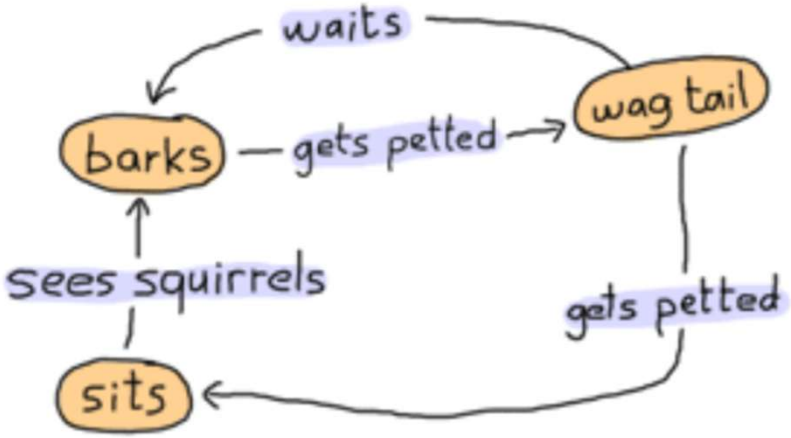
➢ Automat finit: definirea starilor

```
sit() ->
    io:format("Dog is sitting. Goooooood boy!~n"),
    receive
        squirrel -> bark();

        _ ->  io:format("Dog is confused~n"),
              sit()
    end.
```



➢ Implementarea unui automat finit

```
1> c(dog_fsm).
{ok,dog_fsm}
2> Pid=dog_fsm:start().
Dog says: BARK! BARK!
<0.63.0>
Dog says: BARK! BARK!
Dog says: BARK! BARK!
Dog says: BARK! BARK!
3> dog_fsm:pet(Pid).
Dog wags its tail
pet
4> dog_fsm:pet(Pid).
Dog is sitting. Gooooood boy!
pet
5> dog_fsm:squirrel(Pid).
Dog says: BARK! BARK!
squirrel
Dog says: BARK! BARK!
Dog says: BARK! BARK!
Dog says: BARK! BARK!
Dog says: BARK! BARK!
```



http://learnyousomeerlang.com/finite-state-machines#what-are-they

➢ Schimb de mesaje cu transmiterea starii (message passing with data storage)

- Procesul (serverul) este un frigider care accepta doua tipuri de comenzi
  - depoziteaza alimente,
  - scoate alimente .

- Acelasi aliment poate fi depozitat de mai multe ori si poate fi scos de cate ori a fost depozitat.

- La fiecare moment trebuie sa stim ce alimente se gasesc in frigider (starea procesului).

- Starea procesului se transmite prin parametrii functiilor.

kitchen.erl
http://learnyousomeerlang.com/

- Schimb de mesaje cu transmiterea starii

```
fridgef(FoodList) ->
                receive
               % comanda store
               % comanda take
               ....
                end.
```

```
store(Pid, Food) ->
        Pid ! {self(), {store, Food}},
        receive
                {Pid, Msg} -> Msg
        end.
```

```
take(Pid, Food) ->
        Pid ! {self(), {take, Food}},
        receive
                {Pid, Msg} -> Msg
        end.
```

kitchen.erl
http://learnyousomeerlang.com/

➢ Mesaje cu transmiterea starii

kitchen.erl
http://learnyousomeerlang.com/

```
fridgef(FoodList) ->
        receive
            {From, {store, Food}}  ->   From ! {self(), ok},
                                        fridgef([Food|FoodList]);

            {From, {take, Food}} ->  case lists:member(Food, FoodList) of
                                        true ->  From ! {self(), {ok, Food}},
                                                 fridgef(lists:delete(Food, FoodList));
                                        false ->   From ! {self(), not_found},
                                                   fridgef(FoodList)
                                     end;
             terminate   ->     ok
        end.
```

```
6> c(kitchen).
{ok,kitchen}
7> Fridge = kitchen:start([milk, cheese, ham]).
<0.99.0>
8> kitchen:store(Fridge, juice).
ok
9> kitchen:take(Fridge, milk).
{ok,milk}
10> kitchen:take(Fridge, juice).
{ok,juice}
11> kitchen:take(Fridge, juice).
not_found
```

- Varianta: registered process, comenzile **show** (pentru a vizualiza starea) si **terminate**

```
start(FoodList) -> register(fridge, spawn(fun()-> fridgef(FoodList) end)).
```

```
store(Food) ->
   fridge! {self(), {store, Food}},
   receive
      {fridge, Msg} -> Msg
   end.

take( Food) ->
   fridge ! {self(), {take, Food}},
   receive
      {fridge, Msg} -> Msg
   end.
```

```
show() ->
   fridge ! {self(), show},
   receive
      {fridge, Msg} -> Msg
   end.

terminate() ->
      fridge ! {self(), terminate},
      receive
        {fridge, Msg} -> Msg
      end.
```

- Varianta: registered process, comenzile **show** (pentru a vizualiza starea) si **terminate**

```
fridgef(FoodList) ->
   receive
      {From, {store, Food}} -> From ! {fridge, ok},
                                fridgef([Food|FoodList]);
      {From, {take, Food}} ->
                  case lists:member(Food, FoodList) of
                     true ->   From ! {fridge, {ok, Food}},
                               fridgef(lists:delete(Food, FoodList))
                     false ->  From ! {fridge, not_found},
                               fridgef(FoodList)
                   end;
      {From,show} ->  From ! {fridge, FoodList},
                      fridgef(FoodList);
      {From,terminate} -> From ! {fridge, done}
    end.
```

fara apel recursiv

mykitchen.erl

```
2> c(mykitchen).
{ok,mykitchen}
3> mykitchen:start([milk, apple]).
true
4> mykitchen:take(milk).
{ok,milk}
5> mykitchen:store(orange).
ok
6> mykitchen:show().
[orange,apple]
7> mykitchen:terminate().
done
```

mykitchen3.erl

```
fridgef(FoodList) ->
 receive
      {From, {store, Food}} -> From ! {fridge, ok},
                                fridgef([Food|FoodList]);
      {From, {take, Food}} ->
                 case lists:member(Food, FoodList) of
                   true ->   From ! {fridge, {ok, Food}},
                             fridgef(lists:delete(Food, FoodList));
                   false ->  From ! {fridge, not_found},
                             fridgef(FoodList)
                  end;
      {From,show} ->  From ! {fridge, FoodList},
                      fridgef(FoodList)
  end,
  io:format("al doilea receive~n"),
  receive
      {From,terminate} -> From ! {fridge, done}
   end.
```

```
1> c(mykitchen3).
{ok,mykitchen3}
2> mykitchen3:start([]).
true
3> mykitchen3:store(apple).
ok
4> mykitchen3:terminate().
```

procesul este **blocat pentru ca nu poate iesi din primul receive**

Ctrl-G deblocheaza shell-ul
f() dezleaga variabilele

```
User switch command
 --> i
 --> s
 --> c
Eshell V8.3  (abort with ^G)
1> f().
ok
```

http://erlang/doc/man/shell.html

➢ receive … after … end

```
fridgef(FoodList) ->
 receive
      {From, {store, Food}} -> From ! {fridge, ok},
                                fridgef([Food|FoodList]);
      {From, {take, Food}} ->
                 case lists:member(Food, FoodList) of
                   true ->   From ! {fridge, {ok, Food}},
                             fridgef(lists:delete(Food, FoodList));
                   false ->  From ! {fridge, not_found},
                             fridgef(FoodList)
                  end;
      {From,show} ->  From ! {fridge, FoodList},
                      fridgef(FoodList)

  after 30000 ->   timeout
  end,
   io:format("al doilea receive~n"),
   receive
      {From,terminate} -> From ! {fridge, done}
    end.
```

```
1> c(mykitchen3).
{ok,mykitchen3}
2> mykitchen3:start([]).
true
3> mykitchen3:terminate().
al doilea receive
done
```

dupa 30 sec

- in apelul **terminate()**, procesul shell asteapta mesaj de la fridge pentru **receive** din **terminate()**

```
terminate() ->
      fridge ! {self(), terminate},
      receive
        {fridge, Msg} -> Msg
      end.
```

- apelul functiei **terminate()** se incheie numai dupa ce trec cele 30 sec si poate fi prelucrat mesajul "terminate" in al doilea **receive** din **fridgef()**

- receive … after … end

```
fridgef(FoodList) ->
 receive
      {From, {store, Food}} -> From ! {fridge, ok},
                                fridgef([Food|FoodList]);
      {From, {take, Food}} ->
                 case lists:member(Food, FoodList) of
                   true ->   From ! {fridge, {ok, Food}},
                             fridgef(lists:delete(Food, FoodList));
                   false ->  From ! {fridge, not_found},
                             fridgef(FoodList)
                  end;
       {From,show} ->  From ! {fridge, FoodList},
                       fridgef(FoodList);
      {From,terminate} -> From ! {fridge, done}
  after 30000 ->   timeout
 end,
receive
   gata -> io:format("Sunt gata~n")
end.
```

```
gata() -> fridge ! gata
```

```
1> c(mykitchen4).
{ok,mykitchen4}
2> mykitchen4:start([apple]).
true
3> mykitchen4:gata().
gata
4> mykitchen4:show().
[apple]
5> mykitchen4:store(water).
ok
6> mykitchen4:show().
[water,apple]
Sunt gata
```

- functia gata() intoarce imediat, shell-ul nu ramane blocat
- se pot trimite mesaje serverului
- mesajul **gata** este prelucrat dupa ce au trec 30 sec fara o comanda prelucrata de primul receive